Semantic Resume Retrieval for Intelligent Recruitment: A Systematic Review

Dev Pathak[1][0009-0001-7780-3691], Poojan Brahmbhatt[2][0009-0009-4390-9293], Jayneel Brahmbhatt[3][0009-0006-6858-9167] and Parth Goel[4][0000-0002-0074-622X]

[1] University of Massachusetts, Amherst, Massachusetts 01002, USA
[2] University of Southern California, Los Angeles, California, USA
[3] Charotar University of Science and Technology, Changa, India
[4] Charotar University of Science and Technology, Changa, India
lncs@springer.com

Abstract. Keyword based Applicant Tracking Systems have dominated automatic resume assessment for a very long time, yet there is still a gap between these systems and what recruiters actually need. This gap exists because these systems cannot capture the semantic relationships between candidate profiles and job requirements. Recent advances in transformer-based sentence embedding methods have provided a more reliable alternative approach that captures the semantic relationships between resumes and job descriptions. Despite these advances, a systematic and reproducible comparison of different embedding models and retrieval strategies under real-world constraints has been lacking. Therefore, this paper presents a survey of how automatic resume retrieval has evolved from rule-based filtering to semantic based filtering. It also presents an experimental evaluation that compares three sentence-transformer models, MiniLM, MPNet and BGE, using cosine similarity and hybrid retrieval strategies on a dataset of 2,483 resumes across 24 different job categories. The proposed system consists of a pipeline that includes chunk-based document representations, FAISS vector indexing and a two-stage aggregation process. Further, the system is evaluated using Precision@5, Recall@5, Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain at 5 (NDCG@5). After evaluation, it can be observed that smallest models do behave like they are the smallest and hybrid strategies don't benefit all models equally. The results also show that retrieval performance varies a lot depending on the different job categories because they cannot be fully captured by aggregate evaluation metrics. Thus, the findings suggest that the choice of embedding model and retrieval strategy for semantic resume retrieval depends on the application domain.

Keywords: Semantic Resume Retrieval, Sentence Transformers, FAISS vector search, Recruitment Automation, Hybrid Retrieval Scoring

1 Introduction

Hiring the right candidate for a job position has always been a very difficult challenge for organizations. However, with time the task of selecting the best candidate for a position has become more complex due the increasing scale of applications. A single job posting attracts thousands of applications, making it practically impossible for recruiters to review each resume with proper attention. In order to handle these many applications, organizations have started adopting Applicant Tracking Systems (ATS). ATS automates the initial resume screening stage of recruitment, making it easy for the recruiting team [1].

Even though these systems have helped a lot by bringing more efficiency to the recruitment process, they still possess flaws that have persisted for years. The flaw of the traditional ATS systems is that they heavily rely on matching the keywords in the resumes with those in the job description instead of comparing their semantic relationships. The keywords matching works great when the candidates use the exact same terms used by the recruiters in their postings, but it fails in many common situations. One of such situations is when a resume mentions “NLP”, while the job description mentions “Natural Language Processing”. Now here, “NLP” and “Natural Language Processing” are the same skills, but the keyword based system fails to match these two terms. As a result of this, the experienced candidate may be filtered out simply because their resume uses different vocabulary than the job description. For a process where we have to find the best person for a post, these types of strategies tend to exclude most qualified candidates [2].

The main issue with the keyword based ATS is that it matches the words directly instead of matching their meaning. For efficient retrieval, organizations require a system that identifies the right candidate whose skills and experience are relevant to the role, regardless of what they have mentioned in their resumes. The recent advances in natural language processing have made this approach possible.

Transformer based language models designed to generate sentence level embeddings show strong ability to capture the semantic meaning behind the text rather than just its surface form [3]. In our case, these models can be used to convert the resume and the job descriptions into dense vectors and then measuring the similarity between these vectors within a shared semantic space. By doing so, candidates with genuine experience aligned with the position will naturally have embeddings that are closer to the job description, even when the vocabulary used is different. This approach is known as semantic retrieval and it offers a fundamentally more meaningful way to match more candidates with opportunities [4].

Despite the curiosity in applying these techniques to recruitment, there still remains a gap in the research. Most of the current existing studies either evaluate a single embedding model using a publicly unavailable dataset that cannot be reproduced or focus on classifying rather than retrieval. Therefore, there is a shortage of systematic, open-source work that can compare embedding models and retrieval strategies under reproducible conditions where computational resources are limited [5].

This paper addresses these gaps directly. We present a survey on how the automatic resume screening systems have evolved from traditional keyword based ATS to modern semantic based ATS. To support our analysis, we have also provided the experimental results. We build a complete semantic resume retrieval system using publicly available dataset and using open-source embedding models, MiniLM [6], MPNet [7] and BGE [8], along with two retrieval strategy, cosine similarity based retrieval using FAISS vector index and a hybrid strategy that combines both semantic similarity with the keyword based scoring. We have further evaluated the system using standard information retrieval metrics, Precision@5, Recall@5, Mean Reciprocal Rank (MRR) and Normalized Discounted Cumulative Gain (NDCG). These metrics helps in providing a clear, more comparable picture of how each model and strategy performs across 24 different job categories.

The remainder of this paper is as follows. Section 2 reviews related work on automatic resume screening systems and semantic retrieval. Section 3 describes the methodologies we have used, including the dataset, preprocessing pipeline, embedding models, retrieval strategies and evaluation metrics. Section 4 presents the results of our experiments. And Section 5 concludes the paper.

2 Literature Review

The automation of resume screening has undergone significant transformation over the past two decades, initially starting from rule based filtering to the current semantic based retrieval.

The first and earliest screening systems were built based on keyword matching and frequency based scoring. These systems used to work by directly comparing the words present in the resumes with the words present in the job description, assigning higher scores to resumes that contained more words. The most commonly used comparing method was BM25, a probabilistic retrieval model that calculates the frequency of words against the length of the document and produces a score based on that [9]. Although these keyword matching approaches seemed as a practical way to handle large scale applications, they just operated entirely on the surface level of language. All the Applicant Tracking Systems (ATS) built on this kind of bases were efficient in reducing the volume of applications. But they also regularly filter out qualified candidates whose vocabulary did not precisely match with the description [1]. The core limitation of these systems was that they were not capable of understanding the semantic meaning of the words being used.

For handling these issues in the keyword based screening approaches, researchers started using different machine learning techniques for the process. The new methods learned structured representations of resumes and job descriptions to make more meaningful comparisons. Researchers used Named Entity Recognition (NER) as the first method to handle the problem, where resumes are divided into structured sections such as skills, education and experience making it very easy to compare [10]. Another initial research included a neural approach where the resume and job description content were converted into word embeddings that captured some semantic meaning but not at the document level. Qin et al. proposed a semantic aware neural network that worked pretty well for person-job matching using hierarchical attention [12]. Further, Bian et al. extended this approach by adding a multiview coteaching network that combined both texts based and relation based matching and improved the robustness of the models on limited training data [11]. Even after that, these approaches still depended on task specific architectures trained on private datasets, making it difficult to reproduce the results.

The resume screening methods faced a critical turning point in 2019 with the introduction of the BERT model by Devlin et al. [14]. BERT has a bidirectional transformer architecture that helps in understanding the context of text at the sentence and document levels. Researchers quickly started applying the BERT model to recruitment systems. Bhatia et al. used the BERT model's next word prediction ability in order to predict if the resumes are possible continuation of the job descriptions, demonstrating efficient improvements over older approaches [2]. Lavi et al. introduced the conSultantBERT, a Siamese Sentence-BERT model that was fine-tuned specially on the recruitment data, showing that domain based sentence embeddings could further improve the alignment between job descriptions and candidate profiles [4]. Wang et al. explored combining sentence level embeddings with graph based subject term representations in order to capture the structural content of resumes and found that richer document representations led to better predictions [13].

This scalability challenge motivated the researchers to move from using the cross-encoder architectures to using bi-encoder architectures and vector based semantic retrieval. So, in this approach what we do is that instead of encoding both job descriptions and resumes, we convert both the job description and the resumes into dense vector representations that can be easily compared using any similarity search technique. Reimers and Gurevych introduced Sentence-BERT as the foundational framework for this approach, their framework showed that siamese network training could produce sentence embeddings and these embedding can be compared based on their geometric location in the vector space [3]. As these vectors were independently generated it became easy to compare these vectors using similarity search over large collections resumes in the vector database such as FAISS. FAISS supports efficient approximate nearest neighbour search over billions of vectors [15]. This architecture applied to recruitment procedures enables a system to pre-encode an entire resume database and then retrieve the most semantically accurate candidate for any new job description in milliseconds. Rezaeipourfarsangi and Milios demonstrated that treating job and resume matching as a document ranking problem with deep neural retrieval strategy outperformed keyword based systems in ranking quality [5]. More recently, Bevara et al. proposed Resume2Vec, comparing multiple transformer based encoders like BERT, RoBERTa and DistilBERT for resume screening and showed 15.85% NDCG improvement over traditional ATS [16]. Yu et al. took this approach further with ConFit, they applied contrastive learning with dense retrieval encoders to best candidate retrieval and achieved strong performance across both ranking and classification tasks on public datasets [17].

Even after all this progress there still remains a few gaps in the literature, most of the existing studies either evaluate a specific embedding model by comparing it with other options on a private dataset or focus on training task specialized models. There is no systematic open-source benchmark that directly compares multiple sentence transformer models across different retrieval strategies on a publicly available dataset using standard information retrieval metrics. So, this paper addresses that gap by providing exactly such a comparison, evaluating three widely used models, MiniLM [6], MPNet

[7] and BGE [8], across two strategies, cosine similarity and hybrid retrieval, on a publicly available dataset of 2,483 resumes spanning 24 different job categories.

3 Methodology

This section describes the complete experimental framework we used in order to evaluate the semantic resume retrieval. The pipeline includes data preparation, creating custom job descriptions based on the 24 different job categories, text processing, chunking resumes, embedding generation, retrieval strategies and evaluation metrics. Each of the components is described alongside the reasoning behind the design choices made.

3.1 Dataset

The dataset that we have used in this study consists of resumes originally sourced from LiveCareer.com, a publicly accessible resume building platform and is made publicly available as a structured dataset on Kaggle by Anbhawal [18]. This source was selected as the data source because it provides professionally authored resume examples that are well-organized according to their occupational category, making it well suited for evaluating category aware retrieval systems. The dataset contains 2,484 resumes spanning 24 occupational categories, including HR, Information Technology, Engineering, Sales, Healthcare, Finance and many others, covering a diverse range of 24 different professional domains.

The category distribution across the dataset is shown in Table 1.

Table 1. Distribution of resumes across the 24 job categories.

Category	Count	Category	Count
Information Technology	120	Public Relations	111
Business Development	120	HR	110
Advocate	118	Designer	107
Chef	118	Arts	103
Finance	118	Teacher	102
Engineering	118	Apparel	97
Accountant	118	Digital Media	96
Fitness	117	Agriculture	63
Aviation	117	Automobile	36
Sales	116	BPO	22
Healthcare	115	Construction	112

Category labels are used as the relevance ground truth throughout this study. A resume is considered fit to a job description if the assigned category matches the category of that job description. This approach is a widely used evaluation strategy in retrieval research when manually annotated relevance judgments are unavailable [5]. Further, the dataset also contains few labels noise where a subset of resumes doesn’t perfectly align with their assigned category. After manually inspecting the data, it is determined that these noises are not uniform across categories, with some categories such as HR showing strong label consistency, while others such as Chef showing very few amounts of misaligned categories. This characteristic is acknowledged as a known limitation of our evaluation setup and is discussed further in Section 5.

3.2 Job Description Construction

For our experiments, we have manually constructed a set of 24 job descriptions, one for each of the 24 job categories that are present in the dataset. Instead of searching for a secondary dataset to get job descriptions, we decided to manually write job descriptions that reflect the skills, responsibilities and terminologies of each job position for every domain. This approach ensures that we have a clean and controlled relevance mapping between each job description and its corresponding category. Each job description was created based on the vocabulary observed in the actual resumes belonging to that category. By doing this, we have maintained realistic semantic alignment rather than relying on basic job descriptions.

3.3 Preprocessing and Chunking

The text extracted from the raw resumes possesses a lot of noise and it is essential to remove it before creating embeddings. The data cleaning procedure involved removing noise introduced while scraping, replacing non-breaking spaces and other unicode whitespaces with standard spaces and converting multiple consecutive whitespaces and newlines into a single whitespace. No aggressive rewriting or content removal was applied on the data to preserve its semantic meaning.

One of the main decisions we made in our preprocessing pipeline was to use a chunking strategy instead of using fixed length inputs. The transformer based models have input length limitations and can only process inputs of a fixed length, typically between 256 and 512 tokens, depending on the model being used. So, a basic strategy to address this limitation is to use only a fixed length portion of a resume as input. However, the resume data we use has an average length of 811 words, with some resumes exceeding 5,000 words. As a result, what the model does is that it only considers the first 256 or 512 tokens of the resume as input and ignores the remaining section. For example, the skills section, one of the most important sections of a resume, gets ignored as it appears later in the resume. To overcome this limitation, we use a chunking strategy that inputs the resumes in the form of chunks along with the metadata. This enables the models to process every section of each resume, ensuring that no part remains unprocessed.

The metadata that each chunk inherits from its parent resume includes a unique resume identifier, a chunking position index and the resume's category label. All this data is used during the retrieval stage to aggregate the chunks back to their parent resume. Because after chunking the resumes, the chunk collection contains 11,357 chunks across 2,483 resumes, that averages 4.57 chunks per resume and a maximum of 26 chunks for the longest resume in the dataset.

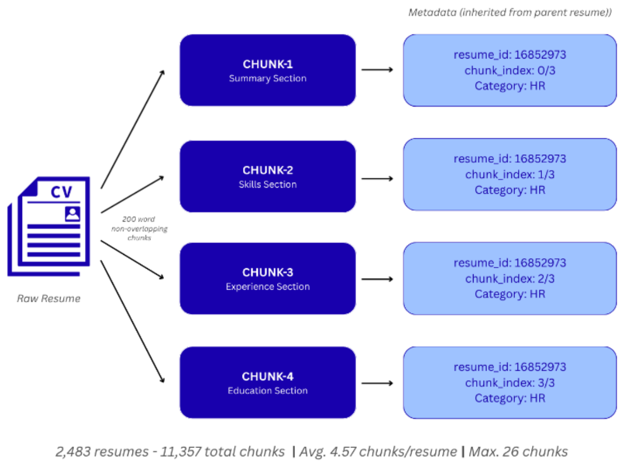

Fig. 1. Illustration of the chunking process: A single resume is split into sequential chunks with fixed length of 200 words, each carrying resume level metadata such as *resume_id, chunk_index* and *category*.

3.4 Embedding Models

As discussed earlier, transformer based models were selected for our experiments because of their biencoder architecture that encodes the whole document independently in fixed dimensional vectors, which makes our retrieval procedure computationally efficient. And when a new job description query is passed, the system only requires a single pass and a simple similarity search in order to retrieve the best corresponding resume. This property makes transformer based a great choice for scalable retrieval systems [19].

Three sentence transformer models were used for this study, based on model sizes, training approaches and embedding dimensionalities.

MiniLM (all-MiniLM-L6-v2): It is a lightweight embedding model that produces 384-dimensional embeddings and was trained through the knowledge distillation from a larger teacher model [6]. The reason for selecting this model was that it provides fast inference with low memory usage, making it suitable for deployment in environments with limited resources and no GPU access.

MPNet (all-mpnet-base-v2): It is an embedding model that produces 768-dimensional embeddings and was trained using a method that helps it understand text by predicting the missing words while understanding the meaning of the entire sentence [7]. The reason for selecting this model was that it is more powerful than the basic sentence encoder and also provides a strong baseline for comparison alongside other models.

BGE (BAAI/bge-base-en-v1.5): It also is an embedding model that produces 768-dimensional embeddings and was specially trained using techniques designed to improve its performance on text retrieval tasks [8]. The reason for selecting this model is that it uses a different encoding approach, where queries are given an instruction prefix, while documents are encoded without one.

All of these three models are openly available through HuggingFace and require no API access or licensing costs that helps with the full reproducibility of our experimental setup. Further, the embedding generation process was performed on a standard CPU without GPU acceleration. Table 2 summarizes the key properties of each model.

Table 2. Summary of embedding models used in study.

Model	Embedding Dimensions	Parameters	Type	Inference Time
MiniLM	384	~22M	General Purpose, Distilled	~9 min
MPNet	768	~109M	General Purpose	~79 min
BGE	768	~109M	Retrieval-tuned	~70 min

3.5 Retrieval Strategies

For our experiments, we have used two retrieval strategies: cosine similarity based retrieval using a FAISS vector index and a hybrid approach that combines both semantic similarity and keyword overlap scoring. Further, both these strategies follow the same two stage retrieval and aggregation design.

For every embedding model, first a FAISS index was constructed over all 11,357 chunk embeddings using the IndexFlatIP index type. This index computes the exact inner product similarity between two vectors. As all embeddings are L2 normalized, the cosine similarity between two vectors is mathematically similar to the inner product search between those two vectors. Further, both of the strategies are discussed in detail:

Cosine Similarity Retrieval: Given a job description embedding q and a set of embedded chunks {c1,c2,…,cn}, the cosine similarity between the job description and each chunk is computed by:

$$sim(q, c_i) = \frac{q \cdot c_i}{||q|| \, ||c_i||} \tag{1}$$

Since embeddings are normalized, the equation reduces to the inner product $q \cdot c_i$, which FAISS computes directly and efficiently.

Overall, the retrieval procedure has two stages. In Stage A, the FAISS index is queried with the job description embedding and the top 30 most similar chunks are retrieved. The value 30, was selected based on the average chunk density of 4.57 chunks per resume. Retrieving 30 chunks makes sure that the available options comfortably contain more than 5 distinct resumes before further aggregation. In Stage B, these retrieved chunks are grouped based on their metadata and the chunk level similarity scores are calculated by using sum pooling:

$$score(r) = \sum_{i \in chunks(r)} sim(q, c_i) \tag{2}$$

where *chunks(r)* denote the set of chunks belonging to resume r that appear in the top 30 retrieved chunks. Further, after that the resumes are ranked based on their aggregated score and the top 5 resumes are returned as the final result. Sum pooling was selected because unlike mean or max pooling, sum pooling naturally rewards the resumes that have multiple chunks retrieved rather than relying on a single strong chunk match.

Hybrid Retrieval: In the hybrid retrieval strategy, we combine both the semantic similarity with the keyword overlapping score. We expect that this will further improve the retrieval performance as the keyword matching alone is a useful not sufficient strategy, but combining it with the semantic meaning of the text will retrieve more relevant results. The hybrid score for a chunk can be calculated by:

$$hybrid(q, c_i) \; = \; 0.7 \; \times \; \hat{s}_i{}^{sem} \; + \; 0.3 \; \times \; \hat{s}_i{}^{kw} \tag{3}$$

where $\hat{s}_i{}^{sem}$and $\hat{s}_i{}^{kw}$ are the min-max normalized semantic similarity and keyword overlap scores for chunk i within the vector database of resume chunks, respectively. The keyword overlap score for a chunk is computed as the fraction of meaningful words in the job description and those that appear in the chunk text:

$$\hat{s}_i{}^{kw} \; = \; \frac{|W_q \cap W_{c_i}|}{|W_q|} \tag{4}$$

where W_q is the set of meaningful words in the job description and W_{c_i} is the set of words in chunk i.

Another factor noticed was that it is necessary to normalize the scores before combining them, because the raw semantic similarity scores from the larger models such as BGE and MPNet cluster in a small range between 0.745 and 0.776 while on the other hand for keyword overlapping, the score varies over a wider range between 0.214 to 0.500 for the same query. So, due to this, the component with greater variance dominates the results for a query. Therefore, it is essential to apply min-max normalization on both the scores in order to rescale them both to a comparable scale. And the influence of both components is handled in a ratio of 70-30.

$$\hat{s}_i \; = \; \frac{s_i - min(s)}{max(s) - min(s)} \tag{5}$$

Unlike Cosine Similarity Retrieval for hybrid retrieval, Stage A uses an even wider candidate pool of 100 chunks rather than 30. By using 100, the keyword component gets a fair opportunity to surface chunks that may rank lower on semantic similarity alone but also contribute meaningful keyword overlap. Afterwards, the top 30 chunks based on the hybrid score are selected from this pool of 100 chunks before Stage B aggregation, which proceeds identically to the cosine strategy using sum pooling.

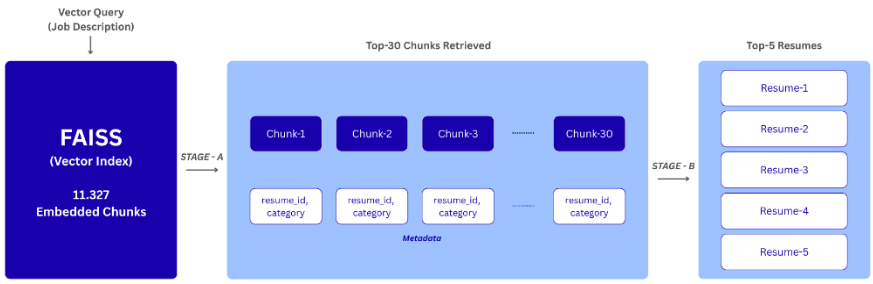

Fig. 2. Architecture diagram of the complete two stage retrieval pipeline: Resume chunks and job descriptions are independently embedded, indexed in FAISS, retrieved as Top N chunks (*Stage A*) and aggregated by resume identifier to produce a final Top 5 resume ranking (*Stage B*).

3.6 Evaluation Metrics

All retrieval results are evaluated at *K=5*, because it replicates a realistic scenario where a recruiter reviews a pool of resumes and then shortlists five candidates per position. And for the evaluation we have used four standard information retrieval metrics that are computed for each of the 24 job descriptions and averaged across all the 24 categories to produce a single score per model and strategy combination. The metrics are:

Precision@5: It measures the fraction of the top 5 retrieved resumes that belong to the same category as the job description, calculated by:

$$P@5 \ = \frac{|\{r \ \epsilon \ Top{-}5: cat(r) = cat\{q\}\}|}{5} \tag{6}$$

Recall@5: It measures the fraction of all relevant resumes in the dataset that were successfully retrieved within the top 5, calculated by:

$$R@5 \ = \frac{|\{r \ \epsilon \ Top{-}5: cat(r) = cat\{q\}\}|}{|\{r \ \epsilon \ D: cat(r) = cat(q)\}|} \tag{7}$$

Mean Reciprocal Rank (MRR): It evaluates how early the first relevant resume appears in the ranked list, calculated by:

$$MRR \ = \frac{1}{|Q|} \sum_{q \ \epsilon \ Q} \frac{1}{rank_q} \tag{8}$$

where $rank_q$ is the position of the first correctly categorized resume in the top 5 list for query q and */Q/=24* is the number of queries.

Normalized Discounted Cumulative Gain (NDCG@5): It measures the ranking quality by assigning higher weight to relevant resumes appearing at earlier positions, calculated by:

$$NDCG@5 = \frac{DCG@5}{IDCG@5}, DCG@5 = \sum_{i=1}^{5} \frac{rel_i}{log_2(i+1)} \quad (9)$$

where $rel_i \in \{0,1\}$ is the relevance of the resume at rank *i* and *IDCG@5* is the DCG of the ideal ranking. NDCG rewards systems that not only retrieve relevant resumes but rank them as high as possible within the Top 5 list.

4 Experimental Results

This section presents the results of our experiments, in which we evaluated three sentence transformer models using two retrieval strategies on an open source dataset. The comparative retrieval performance results for all model and strategy combinations are presented in Table 3.

Table 3. Overall retrieval performance across all embedding models and retrieval strategies.

Strategy	Model	Precision@5	Recall@5	MRR	NDCG@5
Cosine	MiniLM	0.717	0.037	0.824	0.877
Cosine	MPNet	0.683	0.033	0.776	0.788
Cosine	BGE	0.667	0.031	0.788	0.804
Hybrid	MiniLM	0.742	0.039	0.823	0.871
Hybrid	MPNet	0.683	0.034	0.774	0.787
Hybrid	BGE	0.725	0.034	0.774	0.807

After analysing the results, it can be observed that the MiniLM model with hybrid retrieval strategy achieved the highest Precision@5 score of 0.742, which indicates that on an average 3.7 out of every 5 resumes returned for a given job description belonged to the correct job category. The next best performer is again the MiniLM model but with the cosine similarity retrieval strategy, achieving a score of 0.717, making MiniLM model the strongest overall performer out of the three models across both strategies. On the other hand, MPNet and BGE under cosine similarity achieved Precision@5 scores of 0.683 and 0.667 respectively. But the hybrid retrieval strategy shows a different impact across these models. Using the hybrid strategy across the BGE model improved its Precision@5 value from 0.667 to 0.725, while using it across the MPNet model had no meaningful impact, it remained 0.683. Thus, these uneven impacts of the hybrid strategy show difference in how tightly the similarity scores are distributed, which in turn affects how much the keyword overlap component can contribute after score normalization.

The MRR scores across all configurations range between 0.774 and 0.824, which indicates that the first correctly categorized resume consistently appears near the very top of the ranked list regardless of which model or strategy we use. The MiniLM model achieves a MRR score of 0.824, which indicates that on average the first relevant resume appears at approximately rank 1.2, while the lowest MRR value of 0.774 corresponds to approximate rank 1.3. These values indicate that all three models consistently place at least one strong match within the top 5 results, which is very crucial in real life scenarios, as recruiters typically review the highest ranked candidates first. While, the NDCG@5 scores, which additionally reward systems for placing relevant resumes higher within the top 5 ranks, range from 0.787 to 0.877, with MiniLM cosine combination achieving the highest score of 0.877, indicating that MiniLM not only retrieves relevant resumes but also tends to rank them higher within the shortlist as compared to the other models.

The Recall@5 values for all configurations are low, ranging from 0.031 to 0.039. These values are expected because the top 5 retrieval has a structural ceiling that each category in the dataset contains between 22 and 120 resumes. For example, even a perfect retrieval system that identifies the five most relevant resumes would have a maximum Recall@5 of 0.05 when retrieving 5 resumes from a bunch of 100. Therefore, the values observed in our experiments are consistent and indicate a well-performing retrieval system.

Further, examining performance at the job category level reveals a more detailed picture of where semantic retrieval succeeds and where it faces challenges. To further evaluate that, Table 4 shows Precision@5 broken down by occupational category for each model under the cosine retrieval strategy.

Table 4. Per-category Precision@5 for cosine retrieval strategy across all models.

Category	MiniLM	MPNet	BGE
Accountant	0.6	1.0	0.8
Advocate	0.2	0.0	0.8
Agriculture	0.6	1.0	0.2
Apparel	0.6	0.8	0.8
Arts	0.2	0.2	0.2
Automobile	0.2	0.2	0.0
Aviation	1.0	1.0	0.8
Banking	1.0	0.8	0.8
BPO	0.4	0.0	0.2
Business Development	1.0	1.0	1.0
Chef	1.0	1.0	1.0
Construction	1.0	1.0	1.0
Consultant	0.8	1.0	0.6
Designer	1.0	1.0	0.8
Digital Media	0.8	0.6	0.6
Engineering	0.4	0.6	0.8
Finance	0.8	0.8	0.4
Fitness	1.0	0.8	1.0
Healthcare	0.6	0.2	0.8
HR	0.6	1.0	1.0
Information Technology	0.6	0.6	0.8
Public Relations	1.0	0.8	1.0
Sales	0.8	0.2	0.0
Teacher	1.0	0.8	0.6

A clear pattern can be observed from the category level Precision@5 breakdown for each model. Categories such as Business Development, Chef and Construction achieve a perfect Precision@5 score of 1.0 across all three models. This indicates that those categories contain resumes with highly distinctive and consistent vocabulary. Vocabulary that all the three embedding models can easily separate from other content. Similarly strong performance is observed for categories like Aviation, Banking, Designer and Fitness across most model configurations due to the same reasons. While, categories like Arts, Automobile and BPO consistently showcase low precision scores regardless of which model is used. These categories likely contain resumes with broad, spread out vocabulary that overlaps with several other professional domains, making it difficult for our systems to differentiate between them. More importantly, no single model dominates across all 24 job categories, suggesting that retrieval difficulty is more strongly determined by the vocabulary of each occupational category than by the choice of embedding model. Which suggests that lightweight models like MiniLM works well for categories with clear vocabulary, while more ambiguous categories may need additional retrieval techniques to improve accuracy.

Other than that, not only the retrieval quality, the computational cost of embedding generation is also a major practical consideration for real world deployment, especially in environments without GPU accessibility. In our experiments, all embeddings were generated on a standard CPU, where MiniLM required approximately 9 minutes to embed all 11,357 resume chunks, while MPNet and BGE required approximately 79 and 70 minutes respectively. MiniLM's substantially lower inference time, combined with its competitive or leading retrieval performance across most metrics and positions it as a strong candidate for resource constrained deployment scenarios where both accuracy and efficiency matter.

On the other hand, the other two models, MPNet and BGE, both with approximately 109 million parameters, were not able to consistently outperform MiniLM across the evaluation metrics. The results show that for short, domain-specific texts such as resumes and job descriptions, a lightweight distilled model can compete or even outperform large scaled models. In our case, the added size and complexity of larger models did not provide significant im-provements in the performance. A key finding during implementation was that it is essential to normalize the semantic and keyword scores prior to calculating the final scores for this strategy as the keyword component’s highly variable range dominates the results.

5 Conclusion

This paper presented a survey of automatic resume screening systems and an experimental evaluation of the system using transformer based models for intelligent recruitment. A complete open source retrieval pipeline using chunking, FAISS indexing and a two stage retrieval and aggregation strategy was created for the optimal retrieval. In which, we compared three sentence transformer models, MiniLM, MPNet and BGE using two different retrieval strategies, cosine similarity and hybrid retrieval on a publicly available dataset.

After evaluation, the results showed that MiniLM achieved the highest overall retrieval performance despite being the smallest model as compared to the other models. The hybrid retrieval strategy improved performance when score normalization was applied correctly and no single model consistently dominated across all 24 different job categories. Furthermore, the category level vocabulary characteristics were found to be a stronger determinant of retrieval difficulty than the choice of embedding model.

Overall, the findings demonstrated that effective semantic resume retrieval is achievable using open source tools running on standard hardware, making this approach accessible to organizations without large computational resources. The challenges identified during the experiments include label noise, pseudo relevance limitations and category level variability.

After the completion of all the experiments, the limitations of this study should be acknowledged. First, relevance judgments are based on category labels rather than human annotations, which means that some resumes might have in-correct category labels. As a result, resumes that are correctly identified by our system may still be regarded as incorrect during evaluation. Second, the dataset contains a lot of label noise that was introduced during data scraping, which is not uniform across all categories and may introduce inconsistencies in the ground truth. Third, we have assumed and set the value of chunk size as 200 words, that may not align with the natural semantic boundaries within all resume documents. Fourth, the evaluation is conducted on a single publicly available dataset and the generalizability of these findings to larger corpora or other professional domains has not been established.

The future work in this area can be pursued in several different directions. Fi-ne tuning these embedding models on domain specific recruitment data may improve the retrieval performance for the specific categories where general purpose embeddings currently struggle. Replacing the category based labels to human annotated labels would help in providing a more accurate and reliable evaluation framework. Using additional sources of information like GitHub profiles, portfolio links and professional certifications along with the resume text can help in getting more information about the candidate and can lead to better candidate selection. Finally, one of the most common concerns in embedding based recruitment systems is addressing model bias and fairness, as pretrained language models may encode demographic or institutional biases that could unintentionally affect candidate rankings.

References

1. Chavan, M. et al. "Enhancing recruitment efficiency: An advanced Applicant Tracking System (ATS)." ResearchGate, 2024. Available: https://www.researchgate.net/publication/382099903
2. Bhatia, V., Rawat, P., Kumar, A., and Shah, R.R. "End-to-End Resume Parsing and Finding Candidates for a Job Description using BERT." arXiv:1910.03089, 2019. Available: https://arxiv.org/abs/1910.03089

3. Reimers, N. and Gurevych, I. "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks." In Proceedings of EMNLP 2019, pp. 3982-3992. arXiv:1908.10084. Available: https://arxiv.org/abs/1908.10084
4. Lavi, D., Medentsiy, V., and Graus, D. "conSultantBERT: Fine-tuned Siamese Sentence-BERT for Matching Jobs and Job Seekers." arXiv:2109.06501, 2021. Available: https://arxiv.org/abs/2109.06501
5. Rezaeipourfarsangi, S. and Milios, E.E. "AI-Powered Resume-Job Matching: A Document Ranking Approach Using Deep Neural Networks." In Proceedings of ACM DocEng 2023. Available: https://dl.acm.org/doi/10.1145/3573128.3609344
6. Wang, W., Wei, F., Dong, L., Bao, H., Yang, N., and Zhou, M. "MiniLM: Deep Self-Attention Distillation for Task-Agnostic Compression of Pre-Trained Transformers." Advances in Neural Information Processing Systems 33 (NeurIPS 2020), pp. 5776-5788. arXiv:2002.10957. Available: https://arxiv.org/abs/2002.10957
7. Song, K., Tan, X., Qin, T., Lu, J., and Liu, T.Y. "MPNet: Masked and Permuted Pre-training for Language Understanding." Advances in Neural Information Processing Systems 33 (NeurIPS 2020), pp. 16857-16867. arXiv:2004.09297. Available: https://arxiv.org/abs/2004.09297
8. Xiao, S., Liu, Z., Zhang, P., and Muennighoff, N. "C-Pack: Packaged Resources to Advance General Chinese Embedding." arXiv:2309.07597, 2023. Available: https://arxiv.org/abs/2309.07597
9. Robertson, S. and Zaragoza, H. "The Probabilistic Relevance Framework: BM25 and Beyond." Foundations and Trends in Information Retrieval, Vol. 3, No. 4, pp. 333-389, 2009. DOI: 10.1561/1500000019
10. Liang, H. "A Study on the Application of Named Entity Recognition in Resume Parsing." Proc. SPIE 12509, ICHCI 2022. DOI: 10.1117/12.2655918
11. Bian, S. et al. "Learning to Match Jobs with Resumes from Sparse Interaction Data using Multi-View Co-Teaching Network." arXiv:2009.13299, 2020. https://arxiv.org/abs/2009.13299
12. Qin, C. et al. "Enhancing Person-Job Fit for Talent Recruitment: An Ability-aware Neural Network Approach." SIGIR 2018.
13. Wang, Z. et al. "A Deep-Learning-Inspired Person-Job Matching Model Based on Sentence Vectors and Subject-Term Graphs." Complexity, 2021. DOI: 10.1155/2021/6206288
14. Devlin, J., Chang, M., Lee, K. and Toutanova, K. "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding." NAACL-HLT 2019, pp. 4171-4186. https://aclanthology.org/N19-1423/
15. Johnson, J., Douze, M. and Jégou, H. "Billion-Scale Similarity Search with GPUs." IEEE Transactions on Big Data, Vol. 7, No. 3, pp. 535-547, 2019.
16. Bevara, R.V.K. et al. "Resume2Vec: Transforming Applicant Tracking Systems with Intelligent Resume Embeddings for Precise Candidate Matching." Electronics, Vol. 14, No. 4, p. 794, 2025. DOI: 10.3390/electronics14040794
17. Yu, X., Zhang, J. and Yu, Z. "ConFit: Improving Resume-Job Matching using Data Augmentation and Contrastive Learning." arXiv:2401.16349, 2024. https://arxiv.org/abs/2401.16349